

ELBERT MONUMENT, CO, USA

WMO: 722155

Lat: 39.217N

Lon: 104.633W

Elev: 2152

StdP: 78.01

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 03040

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-18.0	-16.0	-22.9	0.6	-12.9	-21.2	0.7	-8.3	12.9	0.4	11.6	2.6	3.0	200	0.657

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.3	30.1	13.0	28.0	12.7	27.3	12.6	16.2	22.9	15.3	22.9	14.6	22.3	4.6	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.9	12.9	18.2	12.7	12.0	17.3	12.3	11.6	16.8	54.0	22.8	50.8	22.9	48.5	21.9	19.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.8	10.4	8.8	DB	-25.0	32.3	2.5	1.7	-26.8	33.5	-28.3	34.5	-29.7	35.5	-31.5	36.7	(4)
(5)			WB	-24.8	17.3	2.5	0.6	-26.7	17.8	-28.1	18.2	-29.6	18.5	-31.4	19.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.7	-1.9	-1.6	2.6	5.5	10.2	16.4	19.3	17.7	14.8	7.6	3.2	-2.3	(6)	
	DBStd	9.07	5.87	5.74	5.36	4.92	4.80	3.81	2.70	2.82	3.96	5.58	5.71	6.20	(7)	
	HDD10.0	1844	368	324	234	146	57	3	0	1	10	109	211	382	(8)	
	HDD18.3	3997	626	557	489	384	253	79	22	44	114	333	456	640	(9)	
	CDD10.0	994	0	1	3	12	63	196	288	239	153	34	6	0	(10)	
	CDD18.3	106	0	0	0	0	1	22	52	24	7	0	0	0	(11)	
	CDH23.3	1736	0	0	0	1	44	438	689	368	184	11	0	0	(12)	
	CDH26.7	398	0	0	0	0	2	118	189	67	21	0	0	0	(13)	
(14) Wind	WSAvg	4.2	4.0	4.3	4.5	5.0	4.4	4.3	3.9	3.8	4.0	4.1	4.2	4.1	(14)	
(15) Precipitation	PrecAvg	520	18	15	39	65	62	58	80	84	36	30	23	17	(15)	
	PrecMax	771	38	28	92	204	166	127	142	154	73	76	82	69	(16)	
	PrecMin	278	6	3	4	2	17	14	26	39	6	2	2	5	(17)	
	PrecStd	107	8	7	26	45	43	30	36	32	20	23	20	15	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.1	17.1	20.0	22.4	27.1	32.3	32.7	30.2	28.9	25.0	20.0	14.8	(19)	
		MCWB	4.0	4.9	6.4	7.4	10.6	12.6	13.4	13.3	12.8	9.9	6.4	3.9	(20)	
	2%	DB	12.0	12.5	17.3	20.1	24.8	29.9	30.2	27.9	27.2	22.4	17.3	12.0	(21)	
		MCWB	2.7	2.7	5.2	7.4	10.1	12.2	13.5	12.8	11.6	8.8	5.2	2.6	(22)	
	5%	DB	8.9	10.2	15.0	17.5	22.4	27.6	28.6	27.3	25.8	20.9	15.1	9.1	(23)	
		MCWB	1.7	2.1	4.3	6.2	9.4	11.7	13.1	12.8	11.2	8.2	4.5	1.6	(24)	
	10%	DB	7.0	7.4	12.4	15.3	20.2	26.1	27.4	25.9	23.9	17.5	12.3	6.4	(25)	
		MCWB	0.4	0.5	3.3	5.6	8.8	11.8	13.5	12.7	11.4	7.1	3.3	-0.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.8	5.0	6.8	9.2	12.3	16.3	18.3	17.7	15.8	11.4	7.0	4.1	(27)	
		MCDB	13.4	15.0	17.3	18.4	22.2	23.8	23.5	22.6	23.0	19.6	17.8	13.0	(28)	
	2%	WB	2.9	3.3	5.7	7.9	11.1	14.9	16.5	15.7	14.1	10.1	5.8	2.9	(29)	
		MCDB	10.4	11.4	15.8	16.9	20.7	22.7	22.9	22.3	21.9	19.3	16.1	10.7	(30)	
	5%	WB	1.7	2.0	4.8	7.0	10.2	13.7	15.6	14.9	12.9	8.9	4.8	1.6	(31)	
		MCDB	8.6	9.4	14.1	16.1	19.2	22.3	23.2	22.0	20.9	18.3	14.3	8.8	(32)	
	10%	WB	0.5	0.7	3.6	5.9	9.5	12.9	14.9	14.1	12.1	7.8	3.6	0.3	(33)	
		MCDB	6.4	6.9	11.7	14.7	17.9	22.4	22.9	21.5	20.4	16.4	11.7	6.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	12.6	12.4	13.2	13.2	13.7	15.4	14.3	14.0	14.8	13.8	13.5	12.5	(35)	
		MCDBR	14.6	15.0	16.5	17.2	17.2	17.7	17.0	16.1	16.8	17.7	16.1	14.9	(36)	
	5% WB	MCWBR	8.1	7.8	7.8	7.7	6.9	6.3	5.0	5.3	6.1	7.7	7.7	8.4	(37)	
		MCWBR	13.8	14.1	15.8	16.2	15.4	15.3	13.7	13.7	15.0	16.0	15.3	14.3	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.224	0.229	0.249	0.272	0.292	0.302	0.326	0.317	0.285	0.253	0.233	0.225	(40)	
		taud	2.647	2.657	2.604	2.547	2.551	2.537	2.509	2.522	2.588	2.667	2.644	2.636	(41)	
	Ebn at Noon		993	1026	1025	1008	985	971	944	948	968	980	971	967	(42)	
		Edh at Noon	64	74	87	99	101	102	104	100	88	72	64	61	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.64	3.59	4.82	5.82	6.47	7.27	6.96	6.28	5.39	4.00	2.90	2.33	(44)	
	RadStd		0.12	0.21	0.35	0.39	0.49	0.32	0.29	0.19	0.26	0.23	0.16	0.12	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (28 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+66	+21

Nomenclature: See separate page